3

Industrial Hygiene Monitoring: Physical Agents

The relative importance of chemical agent evaluation and control typically overshadows the implications of physical agents to the industrial hygiene professional in the semiconductor industry. As will become apparent in this chapter, physical agents are created by equipment and power distribution systems that are found in all modern semiconductor facilities. The other trend becoming apparent is the increased public concern over the potential non-thermal health effects from exposure to electromagnetic fields (EMF). Within the semiconductor industry, this focuses emphasis on fab equipment and computer monitors for extremely low frequency (ELF) emissions and exposures. For this reason, a detailed section on EMF is included in this chapter.

As a background to this section, Fig. 3.1 provides a representation of the electromagnetic spectrum.[1] This spectrum runs from commercial power to the medical x-rays, and includes optical radiation which is discussed in the laser and UV sections, EMF and radio frequency (RF)/ microwave, and ionizing radiation.

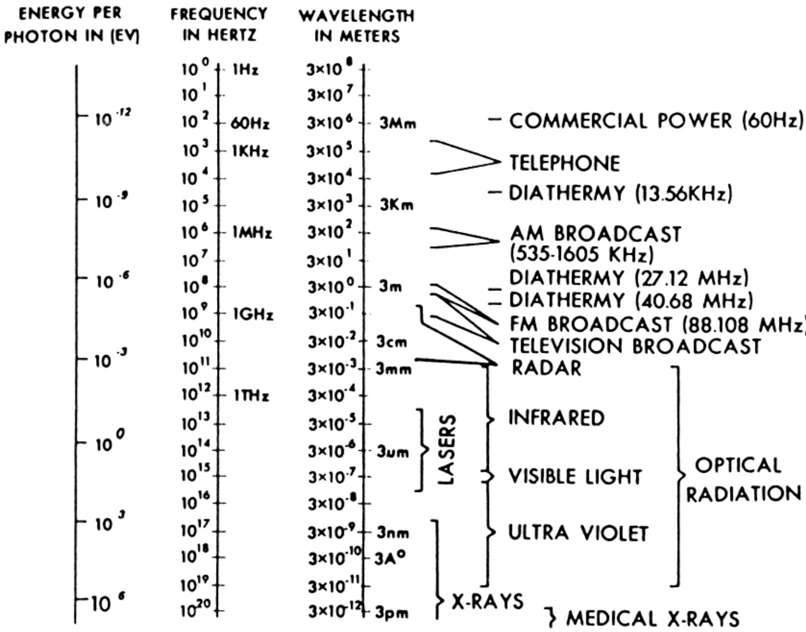

Figure 3.1. The electromagnetic spectrum.[1] *(© 1980, Plenum Press, reprinted with permission.)*

1.0 NOISE

The general exposure of semiconductor employees to noise is not a significant issue within semiconductor manufacturing areas. The major noise sources in cleanrooms are the vertical laminar flow hoods and ceilings, and air escaping from nitrogen wands or feeds on equipment. Background noise from vertical laminar flow hoods and ceilings typically result in noise levels of about 75–78 dBA in the cleanroom. The major noise exposure area within the semiconductor industry is core building areas (or pad areas) where equipment for the operation of the building is located (e.g., HVAC compressors, chillers, boilers, or cooling towers). These areas often run about 85–95+ dBA. Maintenance personnel may or may not work in these areas in durations long enough to trigger a requirement for involvement in a hearing conservation program.

In addition to core building areas, specific diffusion furnaces have been measured up to 86 dBA. With several of these furnaces installed, the noise levels in the service chase area may exceed 90 dBA.

Table 3.1 identifies the recommended duration of exposure versus effective noise level (dBA slow) for permitted exposures using the ACGIH noise standard of 85 dBA as an allowable 8 hr limit.[2] Many semiconductor companies specify 85 dBA or even 80 dBA as the trigger for requiring employees to participate in a hearing conservation program. However, because of the variability of time spent in the core areas, some semiconductor companies designate (and post) building core areas as areas requiring hearing protection and require anyone working in the area, regardless of duration, to participate in a hearing conservation program.

Table 3.1. ACGIH Continuous or Intermittent Noise Threshold Limit Value (93–94).[2]

Duration per Day	Sound Level, dBA
Hours	
24	80
16	82
8	85
4	88
2	91
1	94
Minutes	
30	97
15	100
7.50	103
3.75	106
1.88	109
0.94	112
Seconds	
28.12	115
14.06	118
7.03	121
3.52	124
1.76	127
0.88	130
0.44	133
0.22	136
0.11	139

A prudent rule to follow is that hearing protection should only be considered the final solution when reductions in the noise levels through engineering controls are infeasible or insufficient to reduce levels below 85 dBA.

Some key elements of a hearing conservation program include:

- Baseline and periodic sound surveys, including appropriate postings
- Baseline and annual audiometric exams
- Selection of the proper hearing protection based on the noise levels and employee preferences, with a minimum of three different models of protectors for each category of protector (e.g., ear muffs, foam ear plugs, or molded/plastic ear plugs)
- Training on the elements of a comprehensive hearing conservation program including the anatomy/physiology of hearing, hearing loss, audiometric testing, allowable exposure levels, and the care and use of hearing protection
- Periodic checks on the effectiveness of the program.

The basis of a good hearing conservation program are contained in the cited references.[3]-[6] They should be consulted for additional information.

2.0 NON-IONIZING RADIATION

2.1 Lasers

Lasers primarily present an eye and skin hazard from the direct or reflected energy that is emitted from the *Light Amplification by Stimulated Emission of Radiation* source. The eye is the most vulnerable part of the body because of the light magnification that is created by the lens in the eye and direct contact of the retinal surface with incoming laser energy. Figure 3.2 provides a detailed cross-sectional view of the eye.[7]

Lasers hazards are generally defined by their specific wavelength, power level, and duration/type of emission. The division of the optical spectrum into distinct wavelength regions has been a source of controversy in the scientific community, but the system of the International Commission on Illumination (CIE) is generally accepted. This system forms the basis for the American National Standards Institute (ANSI), Z136.1, *American*

National Standard for the Safe Use of Lasers.[8] Table 3.2 provides three different schemes for dividing the optical spectrum, with the CIE system in the right hand column.[9] This distinction regarding wavelength is important because of the differences in the absorption of the electromagnetic radiation in the eye. Figure 3.3 provides a schematic representation of differences in absorption for electromagnetic radiation in the eye.[10] Figure 3.4 uses the CIE optical spectrum divisions of seven spectral bands to identify eye and skin biological effects.[11]

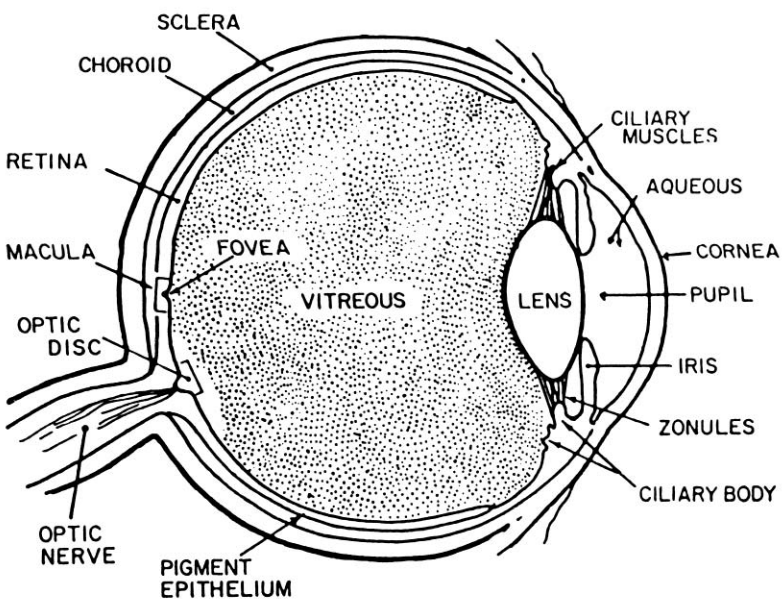

Figure 3.2. The general anatomy of the eye showing the principal structures. [7] *(© 1980, Plenum Press, reprinted with permission.)*

Table 3.2. Different Schemes for Dividing the Optical Spectrum.[9] *(© 1980, Plenum Press, reprinted with permission.)*

Physical #1	Physical #2	Photobiologic (CIE)
Extreme UV (1–10 nm to 100 nm)	Vacuum or extreme UV (1–10 nm to 180 nm)	UV-C (100 nm to 280 nm)
Far UV (200 nm to 300 nm)		UV-B (280 nm to 315–320 nm)
Near UV (300 nm to 400 nm)	Near UV (300 nm to 400 nm)	UV-A (315 nm to 380–400 nm)
Light (380 nm to 760 nm)	Light (400 nm to 700 nm)	Light (380–400 nm to 760–780 nm)
Near IR (760 nm to 4000 nm)	Near IR (700 nm to 1200 nm)	IR-A (760–780 nm to 1400 nm)
Middle IR (4 µm to 14 µm)	Middle IR (1.2 µm to 7 µm)	IR-B (1.4 µm to 3 µm)
Far IR (14 µm to 100 µm)	Far IR (7 µm to 1 mm)	IR-C (3 µm to 1 mm)
Submillimeter (100 µm to 1 mm)		

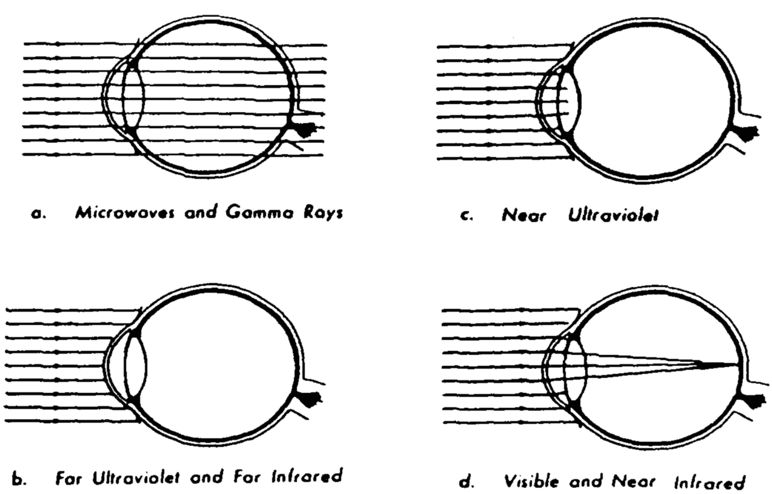

Figure 3.3. Schematic diagram of the absorption of electromagnetic radiation in the eye.[10] *(© 1980, Plenum Press, reprinted with permission.)*

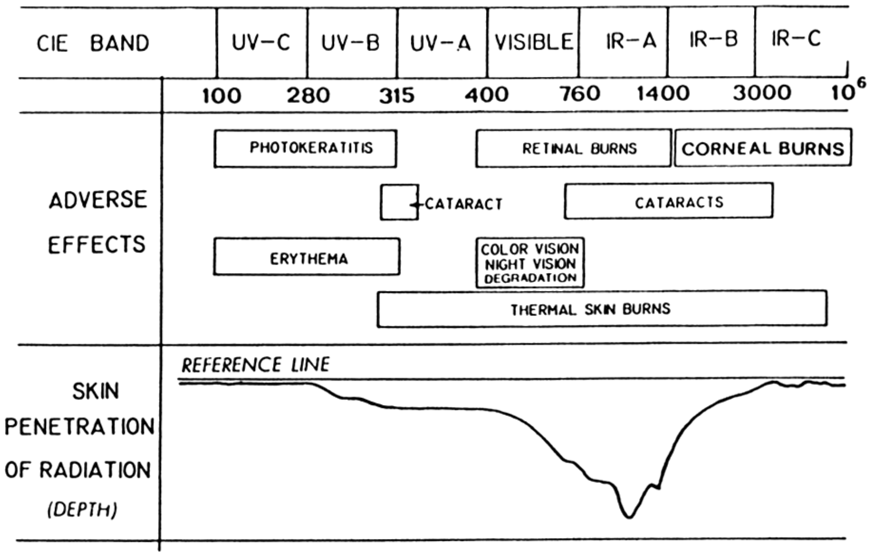

Figure 3.4. The International Commission on Illumination (CIE) seven spectral bands and eye/skin biological effects.[11] *(© 1980, Plenum Press, reprinted with permission.)*

The majority of lasers used in semiconductor processing equipment are low powered (Class 2 or 3A) helium-neon (He-Ne) lasers used for alignment purposes. Typical fab equipment using these lasers includes photolithographic steppers, wafer surface scanners, CVD equipment, and some aligners. Safety precautions needed with these lasers are minimal assuming direct intrabeam viewing is not an issue. The alignment and calibration of the actual laser source itself can create situations where direct intrabeam viewing is a potential hazard.

Most high powered lasers used in the semiconductor industry are: neodymium-yag (Neo:YAG) lasers used for operations such as deburring, trimming leads, mask repair and laser scribing; high powered carbon dioxide (CO_2) laser used for marking the exteriors of both plastic and ceramic IC packages; and deep UV lasers in photolithography. Care should be taken in performing maintenance on laser marking systems, as antimony trioxide, which is a fire retardant is added to the epoxy plastic packages (and beryllium, if ceramic packages containing this compound have been marked), and there is potential for contamination with visible residues from either of theses two toxic metals. Typically, these high powered lasers are categorized as embedded Class 4 (and designated Class 1 with the enclosure in place) with the primary laser exposure concern occurring during maintenance operations such as beam alignment when the beam needs to be energized with the protective covering removed and the interlocks defeated. Table 3.3 identifies some of the more common lasers used commercially and their respective wavelengths.[12]

Table 3.3. Principal Wavelengths of Common Lasers.[12] *(© 1980, Plenum Press, reprinted with permission.)*

CIE band	Wavelength (nm)	Medium	Typical operation
UV-A	325	He-Cd	CW
UV-A	337	Nitrogen	Pulse train
UV-A	350	Argon	CW
Light	441.6	He-Cd	CW
Light	458, 488, 514.5	Argon	CW
Light	458, 568, 647	Krypton	CW
Light	530	Nd frequency-doubled	Pulsed
Light	632.8	He-Ne	CW
Light	694.3	Ruby	Pulsed
Light	560–640	Rhodamine 6G dye	CW/pulsed
IR-A	850	GaAlAs	Pulse train
IR-A	905	GaAs	Pulse train
IR-A	1060	Nd:glass	Pulsed
IR-A	1064	Nd:YAG	Pulsed
IR-C	5000	CO	CW
IR-C	10,600	CO_2	CW

During these maintenance operations, ideally the room containing the laser should be evacuated except for necessary maintenance technicians, and the doors to the room locked and posted with appropriate laser safety signs. However, high powered lasers used in semiconductor manufacturing are often located in large, open manufacturing areas making it impractical to relocate nonessential personnel during maintenance. For these situations, a temporary control area is established. Normally these control areas consist of laser curtains or welding screens capable of withstanding direct contact with the laser beam. Entrance to the temporary control area is usually through a maze entry that is posted with a warning sign whenever the interlocks for the laser are defeated. Other safety precautions during beam alignment are similar to those required for the operation of an open-beamed Class 4 laser (e.g., training, eye protection, written procedures).

The major controls and administrative procedures applicable to lasers used in semiconductor manufacturing are listed in Table 3.4. A detailed explanation of these precautions and others are contained in the reference, American National Standards Institute (ANSI), Z136.1, *American National Standard for the Safe Use of Lasers*.[8] Most laser safety programs within the U.S. semiconductor industry are based on this standard.

Table 3.4. Key Laser Controls Per ANSI Laser Class.[8]

Controls	Laser Classification					
	1	2a	2	3a	3b	4
Labels	-	x	x	x	x	x
Beam Stop or Attenuator	-	-	-	-		x
Alignment Procedures	-	-	x	x	x	x
Training	-	-			x	x
Interlocks on Housing	o	o	o	o	x	x
Temporary Laser Control Area	o	o	o	o	-	-
Laser Control Area	-	-	-	-	x	x
Limited Beam Path	-	-	-	-	x	x
Eye Protection	-	-	-	-	x	x
Key Control	-	-	-	-		x
Skin Protection	-	-	-	-	x	x
Authorized Personnel	-	-	-	-	x	x
Standard Operating Procedures	-	-	-	-		x
Medical Exams	-	-	-	-	x	x
Activator Warning Systems	-	-	-	-		x
Interlock into Room	-	-	-	-	-	x
Warning System on Room Entry	-	-	-	-	-	x
Emergency Off Button	-	-	-	-	-	x

Legend: x Shall
Should
\- No requirement
o Required if Embedded Class 3b or Class 4 Laser

2.2 EMF

A great deal of uncertainty surrounds the question of adverse health effects from electromagnetic fields (EMF). There is currently no consensus within the scientific community on the subject.[13] Information in this section is based on the best available data at the time this book was written. Many studies are in progress that may help to provide further insights into the question of the health risks associated with exposure to EMF.

Until 1979, there was no evidence associating extremely low frequency (ELF) fields with cancer. That year, Wertheimer and Leeper reported a substantial excess of high-current electrical wiring configurations near the homes of children in Denver, Colorado, who had died of brain cancer and leukemia.[14] It should be noted this association was not based on actual measurements of ELF fields, but rather wiring codes were used as a surrogate indicator of magnetic fields from the power transmissions lines and lines feeding the homes. Building materials effectively shield electric fields, but not the corresponding magnetic fields. Therefore, magnetic fields were considered the exposure of potential biological relevance. Savitz also provided additional information regarding wiring configurations and childhood cancers.[15] Since then, more than 50 epidemiological studies have examined this association.[16][17]

There have been over 40 studies of adult cancers and occupational exposures to ELF. Problems with the methodologies used in these occupational studies prevent firm conclusions from being drawn.[17] These problems are:

- In some studies, the effect-estimates tend to be biased upward by the "healthy worker" phenomenon (i.e., because workers tend to be healthy, their background incidence rate of disease is less than the general population).
- The poor quality of the exposure information in virtually all of the studies makes it difficult to rely on "negative" results from any of them.
- Information on potential confounders such as occupational benzene exposure is often absent or highly limited. Benzene is a known human carcinogen that affects the blood-forming systems resulting in leukemia. ELF has also been associated with excess incidence of leukemia in children living adjacent to overhead power transmission lines.

- The occupational literature is susceptible to selective reporting and various jobs and occupations are grouped together depending on the results obtained.

This section emphasizes magnetic field measurements for three reasons:

- Previous epidemiological studies have focused on exposure to magnetic fields.
- Magnetic field measurements, as contrasted with electrical field measurements, are effectively not subject to proximity effects between the measuring device and either the observer, the source, or other conductive objects. This is particularly a problem in characterizing electric field levels in close proximity to workplace devices.[18]
- Industrial exposure to electric fields is typically lower than exposures to magnetic fields when compared to residential electric field/magnetic field exposure ratios. The reason for lower electric fields in the presence of heavy electrical equipment appeared to be the effective shielding provided by metal.[19]

The exact mechanism of cellular events that may be taking place at a physical and biochemical level from exposure to ELF is unknown, but one hypothesis is represented in the schematic shown in Fig. 3.5.[20]

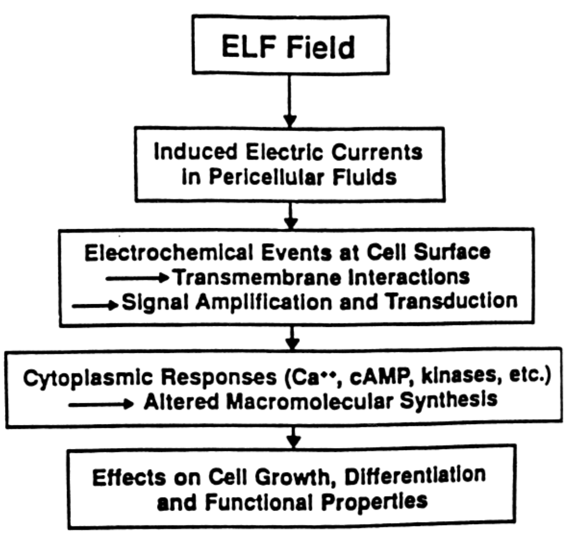

Figure 3.5. Schematic representation of hypothesized cascade of physical and biochemical cellular events triggered by exposure to ELF.[20]

Recognition. Typical workplace exposures to EMF within the semiconductor industry are expected to be far below existing health-based exposure limits (see *EMF Standards & Guidelines for Magnetic Fields).*[18][21][22]

In one study, mean personal exposures of 3.4 milligauss (mG) were found for employees in semiconductor cleanrooms.[21] Table 3.5 provides a summary of the 60 Hz magnetic field exposure measurements from this study.[21] The highest routine cleanroom exposure to ELF in semiconductor manufacturing are expected to be for diffusion furnace operators, where average exposures greater than 5 mG were reported during actual operation of the furnaces. This study also noted that cleanroom personnel whose average measured exposures were noticeably higher than other cleanroom workers were those working in the vicinity of diffusion furnaces. In fact, the study noted this finding was consistent with point measurements reported by Rosenthal and Abdollahzadeh,[18] who found that diffusion furnaces produced proximity readings (2 inches away) as high as 100–150 mG, with the surrounding fields falling off more gradually with distance than other cleanroom equipment studied; even at six feet away from diffusion furnaces, the reported flux densities were 12–20 mG.[21] Figure 3.6 from the same study provides a bar graph representation of the mean time-weighted average (TWA) exposures for cleanroom personnel by process area. The plus symbol (+) shows one standard deviation from the mean, and N = number of workers from whom data was collected.[21] In the same study, the mean level of exposure for maintenance workers was 18.6 mG.[21]

In the recently released comprehensive worker health study that was commissioned by the Semiconductor Industry Association (SIA), the University of California study team looked at physical agents such as radio frequency (RF) radiation, and extremely low frequency magnetic fields (ELF-MF) which arise from the use of electrical current.[22] They defined various ELF exposure parameters based on both fab and non-fab "devices" or equipment that might have a relevant ELF-MF. In the fab category, they included: ion implanters, sputterers, epitaxy reactors, diffusion furnaces, other furnaces, microscopes, fluorescent microscopes, and etchers/ashers. In the nonfab devices category, they included: microscopes, fluorescent microscopes, final test, and probers. Figure 3.7 provides a box plot of the area levels for ELF-MF measured by the researchers during site visits. Figure 3.8 provides similar data for sputterers, etchers, aligners, microscopes, epitaxial reactors, and ion implanters.[22]

Table 3.5. Summary of Semiconductor Cleanroom 60 Hz Magnetic Field Measurements.[21] *(© 1993, ACGIH, reprinted with permission.)*

Cleanroom	Number of Personnel	Mean Time (hr)	Standard Deviation (hr)	B_{mag} Mean (mG)	Standard Deviation (mG)
DDL	16	3.41	0.45	3.85	2.39
MOS V	14	2.97	0.91	2.84	1.15
MOS JIT	5	3.21	0.08	2.09	1.11
BIC II	5	3.43	0.06	3.14	3.10
GaAs:III–IV (Excluding Maint.)	4	6.75	4.55	1.76	1.00
GaAs:III–IV (Including Maint.)	5	6.77	3.94	5.13	7.58
Total (Excluding Maint.)	44	3.55	1.69	3.06	2.00
Total (Including Maint.)	45	3.63	1.74	3.41	3.04

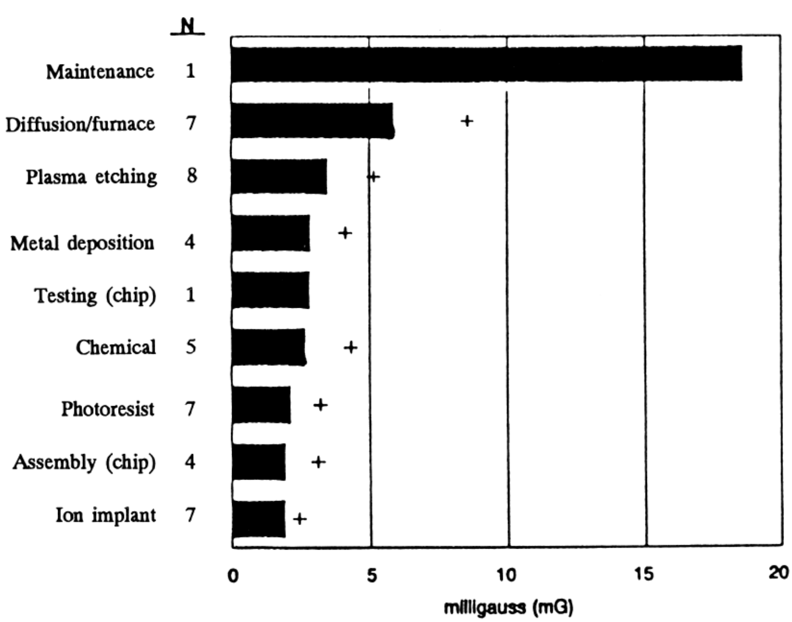

Figure 3.6. Mean exposures of semiconductor workers to 60 Hz magnetic fields during cleanroom processes.[21] *(© 1993, ACGIH, reprinted with permission.)*

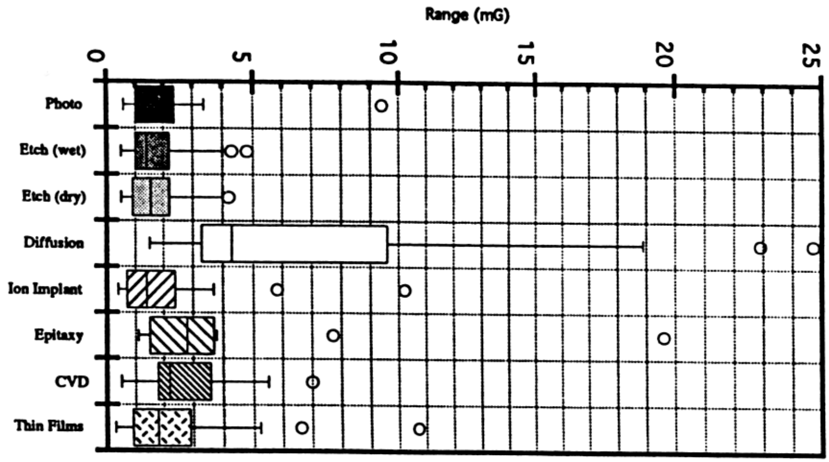

Figure 3.7. Box plot of area ELF-MF measurements for all process groups taken during site visits.[22] (Note: dark lines in each box show median values, the box = the 25th & 75th percentiles, lines extending from boxes = minimum & maximum values, and circles = outliers). (With Permission)

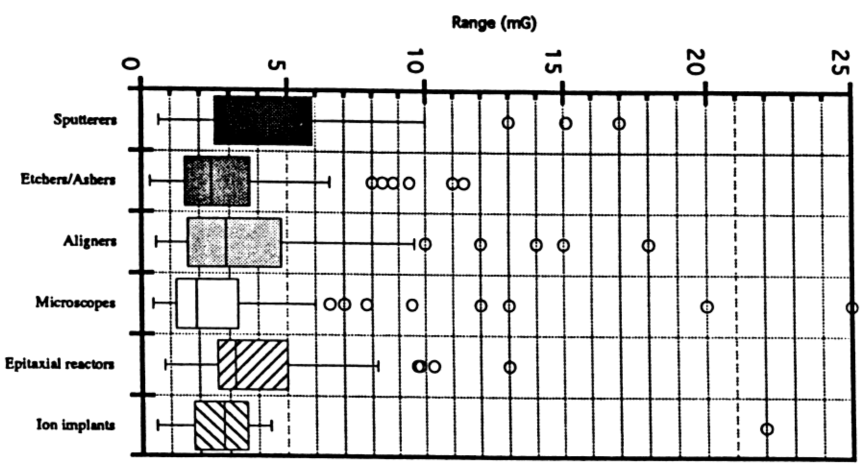

Figure 3.8. Box plot of equipment ELF-MF measurements taken during site visits.[22] (With Permission)

Table 3.6 lists additional emission levels for selected equipment that may be present within the electronics industry, with field strengths that may exceed 5 mG.[23] Table 3.7 provides additional ELF magnetic field readings for selected pieces of equipment and at varying distances from the source.[18]

Table 3.6. ELF Emissions from Selected Equipment in the Electronics Industry.[23]

Equipment	Location	Results (in mG)
Arc Welder, Miller (165 AC)	Operator's Hand	1200
Arc Welder, Miller (310 AC)	Operator's Hand	1500
Blue-M Ovens (N = 3)	Operator's Position	1–9
Cleaner Power Connect Panel, Detrex	—	18
Conveyor Power Supply for Wave Wash	12"	7
Cook Grill	Operator's Position	10
Electrical Panel	12" - 24" - 36" - 48"	100 - 32 - 19 - 6
Grinder, Baldor (N = 2)	Operator's Position	9–15
Hand Sander, Bosch	Operator's Hand	38
Humidifier, Liebert	12" - 24" - 36" - 48"	66 - 31 - 13 - 6
Humidity Testers, Bristol	12"	120
Microscope Power Supply, Unitron	12" - 24"	32 - 3
Microscope Transformer, Techni-Quip	12"	23
Microscope, Unitron	Operator's Hand Position	15
Soldering Iron Power Supply, Weller	12"	24
Soldering Iron Transformer, Weller	12" - 18"	10 - 3
Soldering Iron, Therm-O-Trac	In Operator's Hand	6
Task Light w/Magnifying Glass	Operator Viewing Position	48
Three-Phase Underground Conduit	6" - 12" - 24" - 48"	38 - 19 - 12 - 7

Table 3.7. Magnetic Field Levels Near Devices in Microelectronics Fabrication Rooms.[18]

		Level at 2 inches (mG)		Level at 24 Inches (mG)	
Type	N	Mean	Range	Mean	Range
Aligner	5	13.6	5–20	2.6	1.5–5
Etcher	5	111.2	6–400	4.0	1–7
Sputterer	3	266.7	100–400	6.2	1.5–15
Furnace	4	68.8	5–150	30.8	3–70
Inspection station	13	28.0	5–40	2.7	0.5–5

Overall, the workers exposures to ELF magnetic fields measured in microelectronics cleanrooms were similar to those found in many other occupational settings. Table 3.8 summarizes personal dosimetry measurements on electric utility workers, as reported by the Electric Power Research Institute (EPRI),[24] which gave mean exposure levels for most job tasks ranging between 1–10 mG, while average exposures of over 20 mG were reported for electrical workers involved in distribution and substation tasks. Similar measurements reported for the telecommunications industry[25] gave average worker exposures ranging between 1–10 mG. In summary, the overall average exposures of 3.41 mG found in the Crawford et al. study[21] for workers in microelectronics cleanrooms is consistent with the exposures found in the telecommunications industry and with the lower range of worker exposures in the electrical utility industry. They concluded that this data shows that, as far as magnetic ELF fields are concerned, no unique conditions exist in the cleanroom environment that are not present in other routine electrical occupations or job tasks.[21]

Table 3.8. Mean Extremely Low Frequency Exposure Values for Workers in Electric Utility, Telecommunications, and Semiconductor Industries.[21] *(© 1993, ACGIH, reprinted with permission.)*

Industry – Task	Sample Size	Mean (mG)
Electric Utility[A]		
Generation	881	8.26
Transmission	357	9.06
Distribution	853	25.75
Substation	944	23.81
Electrical	268	4.08
Outdoor	217	2.19
Shop	258	3.93
Office	607	1.78
Nonwork	–	1.49
Telecommunications[B]		
Telecommunications vault	76	8.21
Shared pole	72	4.65
Residence customer premises	188	2.30
Business customer premises	148	3.27
Central office	98	2.23
Work center	138	1.66
Work area protection setup	120	3.24
Travel	572	2.26
Lunch or break	257	1.82
Nonoccupational	29	1.73
Semiconductor[C]		
Cleanrooms	45	3.41

[A]From Electric Power Research Institute Project.[15]
[B]From Rainer et al.[11]
[C]Present study, Table II.

Evaluation. The type of meter used in a ELF survey depends on the nature of the survey to be performed. To respond to typical employee ELF concerns, a 50/60 Hz magnetic field meter measuring root-mean-square (RMS) is adequate. This type of meter does not account for possible harmonics at 50/60 Hz. However, for standard electrical equipment, harmonics are not expected to contribute a significant amount of ELF emissions.[18]

If more accurate ELF measurements are needed, an ELF magnetic field meter that accounts for the harmonics of 50/60 Hz power frequency fields is used. An example of this is the measurement of ELF fields resulting from switch mode power supplies that feed three phase primary distribution lines. The most significant frequency created by these distribution lines is at 150/180 Hz. When three phase primary distribution lines are present, these components can be additive.

For common ELF surveys, it is usually adequate to measure only the magnetic field. However, to thoroughly characterize ELF emissions, a meter capable of measuring both magnetic and electric fields is necessary. Characterizing emissions from an operation or at a workstation are typically adequate for the types of surveys performed within the semiconductor industry. However, if special circumstances arise, a personal dosimeter may be required.

Personal dosimeters measure the person's average and peak exposure during the shift. They are particularly useful when the person works in a variety of areas during the shift and the emphasis is on determining the person's exposure rather than characterizing emissions from ELF sources. If a personal dosimeter survey is performed, it should also include the characterization of non-occupational exposures along with workplace exposures of the subjects.

Video display terminals (VDTs) contain cathode ray tubes (CRTs) that emit both 50/60 Hz ELF fields and higher VLF fields (from 15 kHz to 70 kHz depending on the VDT). Therefore, additional measurements and meters are needed to accurately survey VDT emissions. There are various field properties that impact the accuracy of readings taken on electric (E field) and magnetic (H field) fields associated with a VDT. Figures 3.9 and 3.10 provide a visual representation of the magnetic and electric fields that are typically emitted from a VDT.[25] The important point to note is that the presence of a object (such as the operator at a VDT workstation) will affect the electric field reading. Care must be taken to ensure that individuals taking electric field measurements keep their body out of the E field by using an extended holder for the probe on the monitoring device.

Most reasonably priced ELF survey meters use a coil-type of detector. The majority are single-axis (i.e., they only read the field in a given direction). Having only a single axis requires that the industrial hygienist rotate the detector coil (or the entire instrument if the coil is internal) in all directions until the maximum reading is found. The alternative is to take three readings, one each along the X, Y, and Z axis, and compute the square root of a sum of squares (RMS): Reading = $(X^2 + Y^2 + Z^2)^{½}$.

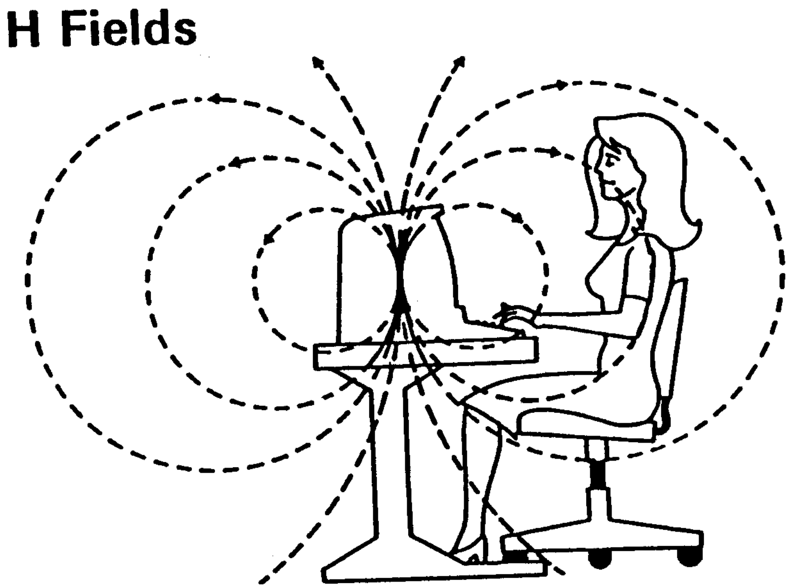

Figure 3.9. VDT magnetic field emissions are unperturbed by the presence of an operator.[26] *(Reprinted with permission.)*

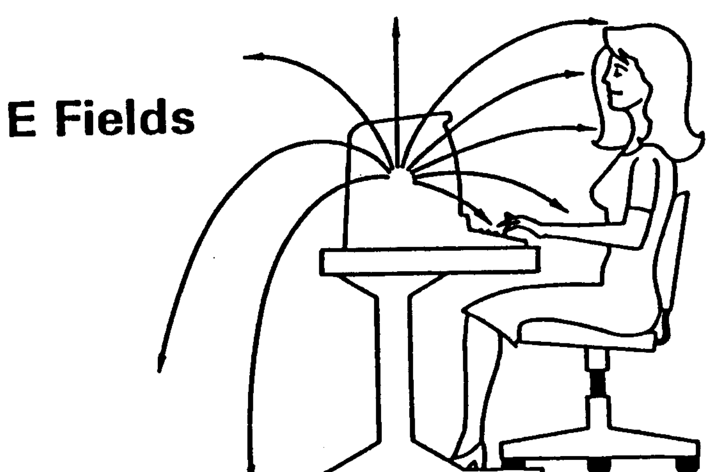

Figure 3.10. VDT electric field lines incident on VDT operators are affected by their body surface.[26] *(Reprinted with permission.)*

As an additional feature in some of the more expensive meters, the manufacturers have designed-in three internal coils and do the summing automatically.

Some ELF meters will give erroneous readings if subjected to radio frequency emissions (e.g., RF from radio transmitters, leaking sputtering units, plasma etchers). In one study, it was found that RF levels > 0.03 mW/cm^2 dramatically changed the reading of an ELF meter.[18] However, this is not a major concern except in rare situations because the background RF levels in fab areas are typically below this level.

ELF Survey Procedures. Because mechanisms of ELF-induced health effects have not been clearly established, the most appropriate parameter for characterizing exposure is not known. In particular, the appropriate averaging time for characterizing ELF exposures is unclear (e.g., should peak or time-weighted averages be used). Spot measurements coupled with estimations of time spent at the work station are considered adequate for typical surveys.

Steps in conducting an ELF survey include:

1. If in doubt, determine the frequencies that are of concern from the equipment or the workstation that is being surveyed. (This may be necessary to ensure the equipment does not produce frequencies higher than the meter is capable of measuring that may cause interference with the meter, or frequencies that may be a greater potential health concern than the ELF. Equipment used in the semiconductor industry that may produce high frequency emissions includes diffusion furnaces, VDTs, sputterers, plasma etchers, plasma ashers, ion implanters, epitaxial reactors, and microwave ovens.)
2. Turn all electrical equipment in the work area off and take background readings (if this is practical).
3. Next turn the electrical equipment in the work area back on and measure the emissions at various locations in the work area at both source and operator positions. (The source readings measure emissions, while the operator position measurements indicate exposure.)
4. Record the measurements on a diagram of the work area.
5. With respect to the source, record the location of the operator at various times in the shift.

There are two emphases for ELF surveys, one is characterizing the emissions from the source and the other is estimating the employee's exposure. When estimating the employee's exposure, the primary focus is measuring exposures to the trunk and/or head. However, if there is significantly more exposure to the hands and feet (i.e., 5x the torso), these data should also be recorded.

EMF Standards and Guidelines for Magnetic Fields.

International. The International Non-Ionizing Radiation Committee (INIRC) of the International Radiation Protection Association (IRPA) recommends a 50/60 Hz magnetic field limit of 5 G for the entire workday, 50 G for exposures of less than 2 hours duration, and 250 G for exposure of the limbs throughout the workday. The IRPA/INIRC exposure limits for members of the general public were set at 20 percent of the occupational limits (i.e., 1 G for continuous exposures and 10 G for exposures during periods of a "few hours per day").[27]

British. The National Radiological Protection Board (NRPB) recommends a magnetic field TLV® of 19 G at 50 Hz and 16 G at 60 Hz. The formula is: limit (in G) = 940/f; where f = frequency (Hz). This is for frequencies from 10 Hz to 750 Hz. At frequencies from 750 Hz to 50 kHz, the exposure limit is 1.25 G. The guideline also states that occupational personnel should not be exposed to the maximum permissible field levels for more than 2 hours per day.[28]

German. German guidelines recommend a magnetic field limit of 50 G at 50 Hz and 46 G at 60 Hz. The formula is: Limit (in G) = 271.36/ f0.4325; where f = frequency (Hz). This is for frequencies from 2 Hz to 10 kHz.[29]

U.S. The ACGIH recommends a magnetic field TLV of 12 G at 50 Hz and 10 G at 60 Hz. The formula is: TLV (in G) = 600/f; where f = frequency (Hz). This is for frequencies from 1 Hz to 30 kHz.[30]

The *World Health Organization (WHO)* report, based on data gathered from human exposure to time-varying magnetic fields, states that, for frequencies of 50/60 Hz:[31]

1. Exposures below 50 G have not been shown to determine any significant biological effect
2. Fields of 50–500 G have determined "minor transient health consequences for short-term exposure (a few hours), but unknown health consequences for long-term exposures (many hours, days, or weeks)"

3. Exposures to fields greater than 500 G overpass cell stimulation levels and may result in health problems

U.S. VDT Emission Limits. The health/safety-based American Conference of Governmental Industrial Hygienists (ACGIH) standard for 1 Hz to 30 kHz frequencies magnetic fields is TLV (in G) = 600/f; where f = frequency in Hz. From 30 kHz to 3 MHz, the magnetic field standard is 20 mG (1.6 A/m or 2.65 A^2/m2). The ACGIH standard for electric fields is 25 kV/m from 0 Hz (DC) to 100 Hz. For frequencies in the range of 100 Hz to 4 kHz the TLV (in V/m) = (2.5 x 106) / f; where f = frequency in Hz. A value of 625 V/m is the exposure limit for frequencies from 4 kHz to 30 kHz. For frequencies from 30 kHz to 3 MHz the TLV is 614 V/m (377,000 V^2/m^2).

Swedish VDT Emission Limits. On October 4, 1991, the Swedish National Board for Measurement and Testing (Statens Matoch Provstyrelse, or MPR) recommended the first guidelines to limit emissions of low frequency (ELF) electromagnetic fields (EMFs) from VDTs. The board called for an ELF magnetic field limit of 2.5 mG at 50 cm from a VDT. The MPR also proposed an ELF electric field limit, as well as revised guidelines for very low frequency (VLF) electric and magnetic fields (see Table 3.9).[32]

Table 3.9. Swedish EMF Guidelines for VDTs MPR 2.[32]

	ELF (5 Hz – 2 kHz)	VLF (2 kHz – 400 kHz)
Magnetic Fields	2.5 mG[(a)] (250 nT)[(b)]	0.25 mG[(a)] (25 nT)[(b)]
Electric Fields	25 V/m[(c)]	2.5 V/m[(d)]

Footnotes:

(a) Magnetic fields are now specified by their average strength (root-mean-square, or RMS), rather than by peak levels.

(b) Measured 50 cm from the VDT in three planes: one around the middle, one 25 cm above the middle, and one 25 cm below it.

(c) Measured 50 cm from the front of the VDT.

(d) Measured 50 cm from all sides of the VDT.

In announcing the new guidelines, the MPR stressed that the emissions limits are voluntary as the Board does not have the authority to regulate occupational health risks. Similarly, the testing of VDT emissions is not mandatory. The MPR noted that the recommended limits are based on what is technically feasible to measure and on what is achievable "today or within the near future." The MPR said that the guidelines are not based on health risks because "there are no proven biological reasons" for limiting VDT EMFs, according to Swedish experts.[32]

Residential Exposures. In many instances it is useful to compare occupational survey monitoring results with typical residential exposure levels. This information provides a reference point for comparison with workplace exposure levels. It is recommended that this information be coupled with relevant health-based occupational exposure limits.

Household magnetic fields are highly variable and change quickly with power demand cycles. Results suggest magnetic field measurements are log normally distributed. Typical resultant mean values in most rooms are about 1 mG, with a standard deviation of about twice the mean, and peak values over 12 mG.[33] Measurements taken during the use of appliances are much higher and have resultant means of about 9–20 mG, with standard deviations of about 40–75 mG and some peak values exceeding 1,000 mG.[33] However, actual residential exposures would probably be considerably less depending on the proximity of the person to the location where the appliances are running. Table 3.10 contains information on ELF emissions from different types of household appliances and common industrial equipment.[34]

Controls. Three ways of reducing ELF exposures are time, shielding, and distance. These control techniques are applicable to any physical agent.

- Reducing the time spent in an ELF field is the simplest method for reducing the exposure to the individual.
- Significant shielding of employees from electric field exposures can be accomplished by containing the source in a metal cabinet. However, shielding is not practical for controlling ELF magnetic fields.
- Increasing the distance between employees and the ELF source is often the only control technique available for reducing exposure. However, from a practical standpoint, because of numerous variables that would go into calculating the drop in the field strength, no general formula can be used to calculate the drop off of magnetic fields with distance from typical industrial ELF sources.

Table 3.10. Magnetic Fields at Distances from Appliance Surfaces.[34]

Appliance	Magnetic Field in Milligauss At 4 inches	At 1 Foot	At 3 Feet
Clothes Dryers	4.8 to 110	1.5 to 29	0.1 to 1
Clothes Washers	23 to 3	0.8 to 3.0	0.2 to 0.48
Coffee Makers	6 to 29	0.9 to 1.2	<0.1
Toasters	10 to 60	0.6 to 7.0	<0.1 to 0.11
Crock Pots	8 to 23	0.8 to 1.3	<0.1
Irons	12 to 45	1.2 to 3.1	0.1 to 0.2
Can Openers	1300 to 4000	31 to 280	0.5 to 7.0
Mixers	58 to 1400	5 to 100	0.15 to 2.0
Blenders	50 to 220	5.2 to 17	0.3 to 1.1
Vacuum Cleaners	230 to 1300	20 to 180	1.2 to 18
Portable Heaters	11 to 280	1.5 to 40	0.1 to 2.5
Exhaust Blowers	3 to 120	2.5 to 37	<0.1 to 3.1
Hair Dryers	3 to 1400	<0.1 to 70	<0.1 to 2.8
Electric Shavers	14 to 1600	0.8 to 90	<0.1 to 3.3
Televisions	4.8 to 100	0.4 to 20	<0.1 to 1.5
Fluorescent Fixtures	40 to 123	2 to 32	<0.1 to 2.8
Fluorescent Desk Lamp	100 to 200	6 to 20	0.2 to 2.1
Saber & Circular Saws	200 to 2100	9 to 210	0.2 to 10
Drills	350 to 500	22 to 31	0.8 to 2.0

The conventional industrial hygiene procedure of reducing exposure to reduce health effects is considered appropriate. However, as has been mentioned previously, the mechanisms of ELF-induced health effects have not been identified. Also, some published studies contradict the idea less exposure is better. These studies indicate there may be window effects, that is, effects of ELF exposure may occur at only certain frequencies and intensities and a dose-response relationship may not exist.

Summary. The following summary information comes primarily from an excellent booklet put out by the Carnegie Mellon University, titled "Electric and Magnetic Fields from 60 Hertz Power: What do we know abut possible health risks?[35]

- 60 Hz (60 cycles per second) electric and magnetic fields come from electric power. They are found around all electrical appliances, house wiring, power lines in the street, and high voltage transmission lines.
- There is clear evidence that 60 Hz fields can produce various hormonal and other changes in living things. It is not yet clear if these changes can result in risks to public health.
- Possible risks of concern include the promotion of cancer (i.e., helping the growth of existing cancer); developmental abnormalities (i.e., birth defects); and various neurological effects such as chronic depression.
- There have been many very good scientific studies of the possible health risks of fields.

Taken together, the results are very complicated. Careful and responsible scientists do not yet agree on whether 60 Hz fields pose a risk to public health and, if they do, how serious that risk might be.

- It is not clear what aspect of 60 Hz fields (if any) poses a risk. There is evidence which suggests that, across the range of field strengths commonly encountered by people, stronger fields may not pose greater risks than weaker fields. This means that the usual assumption that "more is worse" may not be correct for the case of 60 Hz fields. With the scientific evidence that is now available, it is not possible to establish a "safe field" standard.
- There is a great deal of research going on to learn more about possible health risks of 60 Hz fields. New evidence from this research should help to reduce some of the current uncertainty.

2.3 UV

The predominant wavelengths of ultraviolet (UV) light currently used in photomasking are 365 nm or above, but UV lamp spectra do contain significant energy in the actinic region below 315 nm. Normally, the intensity of the UV radiation escaping from the equipment is less than the TLV. Occasionally during maintenance, the alignment of the UV lamp requires them to be energized outside the equipment cabinet or with normal protective filters bypassed. Exposure levels during this operation can exceed the TLV, but standard clean room attire (e.g., smocks, vinyl gloves, face masks, and polycarbonate safety glasses with UV inhibitor) usually is adequate to attenuate the UV light to below the TLV.[36]

While the predominant wavelengths for ultraviolet (UV) lamps used in photolithography are 365 nm or above, the quest for smaller features in advanced ICs are leading to the use of exposure sources with smaller wavelengths such as deep-UV and x-rays. One new technology for this purpose is the use of krypton-fluoride excimer lasers in steppers. These steppers use a wavelength of 248 nm at Class 4 laser power outputs. However, enclosures for these systems result in their classification as Class 1 lasers under normal operation. As with other embedded Class 4 laser systems used in semiconductor manufacturing, the main concern is when interlocks for the system must be defeated during beam alignment. Controls and safety design considerations for these system are contained in Ref. 8.

While a number of sophisticated, artificial UV sources are utilized in the semiconductor industry, the highest exposures to UV light in the actinic region typically occurs outside. On a sunny day in Santa Clara County, California, the effective UV irradiance approximately equals the 15-minute TLV.

2.4 RF/Microwave

Table 3.11 is a list of common equipment used in the semiconductor processing that generates RF or microwave radiation, and the operating frequencies of the equipment. With regard to operator safety, the primary concern for exposure to RF energy comes during plasma etching and ashing.[38][39] Typically, the leakage of RF energy can be caused by *(a)* misaligned doors, *(b)* cracks and holes in the cabinets, *(c)* metal tables and electrical cables acting as antennae due to improper grounding of the etcher, and *(d)* no attenuating screen in the viewing window of the etcher.[39][40]

RF exposure can also occur during the maintenance of etchers, particularly if the equipment cabinet has been removed. An exposure of 12.9 mW/cm^2 was found at the top of a Tegal 915 plasma etcher with the cover removed for maintenance.[40] The actual RF radiation leakage in the area where the operator stands was typically less than 4.9 mW/cm^2. Leakage from the sides and backs on many Perkin-Elmer 4400 Series Sputterer units was found to exceed the TLV.[36] Most of the leakage was attributable to cracks in the cabinets caused by repeated removal of the maintenance panels. In newer Perkin-Elmer models, panels with wire mesh along the seams prevent significant leakage. Series 4400 sputterers can be retrofitted with wire mesh or, alternatively, copper tape can be used to cover the seams to reduce the leakage.

Two excellent references on the subject of exposure standards and measurement of RF and microwave fields are contained in the ANSI/IEEE documents (Refs. 41 and 42).

Table 3.12 identifies RF/Microwave penetration depths in the muscle, skin, and other tissues with water content for different frequencies.

Table 3.11. Common RF/Microwave Radiation Producing Devices used by the Semiconductor Industry.

Equipment	Frequency (MHz)
Sputterers	13.56
Plasma Ashers	13.56 and 2450
Plasma Etchers (most)	13.56
X-ray Lithography Sources	50–55
Mold Preheaters	70–110
Handheld Radios*	460–470
Microwave Ovens	2450

* Handheld radios 7 watts or less are exempt from the ACGIH RF/microwave standard.

Table 3.12. RF/Microwave Penetration in Muscle, Skin, and Other Tissues with High Water Content.[43]

Frequency (MHz)	Depth of Penetration (cm)
1	91
27	14
40	11
433	3.6
915	3.0
2450	1.7
5800	0.72
10,000	0.34

3.0 IONIZING RADIATION

The exposure of semiconductor fabrication area and maintenance personnel to ionizing radiation is quite limited. The largest number of personnel work with x-ray generating equipment, such as the ion implanters and cabinet (package) x-ray units. Use of radioactive material is limited primarily in small test labs doing coating thickness testing (QA/Reliability, Packaging) or through the use of general license anti-static units that may be installed on inert gas lines used for particle cleanup. The most significant quantities of radioactive materials used in semiconductor manufacturing are utilized in the fine leak detection testing of the hermeticity of both plastic and ceramic IC packages with krypton^{85} radioactive gas. Because krypton is a gas, the potential exists for leakage or discharge of the gas into the workroom or release into the atmosphere.

3.1 Radioactive Material

Cabinet x-ray systems are normally used to check the thickness of metal coatings on the IC. Alternatively, the older technology is to measure metal coatings with sealed beta-emitting radiation sources. These sources are typically categorized as "general license" radiation sources in the U.S. from the radiological health department in the applicable state. Such sources typically contain microcurie quantities of radioactive material that are encapsulated (which leads to the term "sealed sources"). Containers for these sealed sources are prominently labeled with an ionizing radiation emblem. This arrangement typically does not present a significant hazard under normal operations. The problem usually comes from either misplacing the source (it is illegal in the U.S. to dispose of the source in normal domestic trash—all sources must be returned to the original manufacturer or disposed of as radiological waste), or from the protective housing being breached due to mishandling.

Regulations require that the sources be "leak tested" periodically (usually on a semi-annual or annual basis) to ensure that they are not leaking the radioactive particulate material that is enclosed. Leak testing involves wiping the source enclosure with a small swab, and submitting the sample for analysis by a certified radiological testing laboratory. Leaking sources are immediately removed from service and returned to the manufacturer. Employees should be notified of the leak test results. The area around the leaking source needs to be surveyed with a Geiger-Mueller (GM) counter, and points of leakage ("hot spots") are wiped clean. The area must be rechecked to ensure all contamination has been removed.

Another source of radioactive material sometimes used in the semiconductor industry are anti-static devices containing polonium^{210}, an alpha emitter. While used extensively in the past, currently, the use of these sources is primarily limited to certain mask making operations. Typically the anti-static devices are leased from the manufacturer for a specific period of time (e.g., 13 months). The supplier of the anti-static device normally performs a periodic wipe test of the source, but the users should maintain copies of the records of these tests. Design problems have occurred in the past with the enclosure housings, and notices of potential leakage of radioactive material conveyed to end users.

The two principal "specific license" radiation sources used in semiconductor manufacturing are krypton^{85} gas (up to 50 curies) used in fine leak testing systems and cobalt^{60} (up to 26,000 curies) used in irradiators for testing the ability of ICs to withstand exposure to gamma radiation in military and space. Under normal operating conditions, personnel exposures are typically less than 500 millirems per year.[36] Controls for these systems include:

- Isolation in rooms with access limited to necessary personnel
- Posted radiation warning signs on the doors to the rooms containing the radioactive sources
- Continuous radiation monitors with alarms (and auto shutdown/isolation for fine-leak systems)
- Dedicated exhaust system and negative pressure room for Kr^{85} systems
- Monitoring exposures with personal dosimetry (e.g., radiation film badges)
- Regular maintenance of alarms and interlocks
- Regular checks for radioactive material leakage
- Safety training for operators and technicians
- Ensuring radiation exposures are kept As Low As Reasonably Achievable (ALARA)

Also, materials that come in contact with Kr^{85} (e.g., exposed ICs, used pump oil, valves, and O-rings) are surveyed to ensure they do not emit excessive levels of radiation (e.g., >2 mR/hr). Detailed information on exposures and controls from Kr^{85} fine leak detection systems are available in Ref. 44.

Small "specific license" alpha sources (e.g., micro- and millicuries of $Americium^{241}$) are used in the failure analysis process. These sources are covered by a thin protective coating called a window that allows alpha particles to be generated from the source to test the integrated circuit's ability to operate when bombarded by alpha particles. The direct wiping of the protective window could damage it. Because of this, the integrity of the window is periodically checked (e.g., semi-annually) by holding the source over a clean sheet of paper, gently tapping the base of the source and wiping the area on the paper under the source. Any detectable leakage usually triggers removal of the source and shipment back to the manufacturer.

Employees are required to be trained on the hazards of the source(s), and an individual must be designated as the Radiation Safety Officer when specific licensed radioactive material is used at the facility. Records are required to be kept on the specifics of the sources at each location (manufacturer, type, quantity, storage and use location, person the sources are assigned to, and leak testing/GM counter results) and a locked/secure location is recommended.

3.2 X-ray Generating Machines

A wide variety of electrical equipment is used in semiconductor manufacturing that has the potential for generating x-rays. This list of equipment includes scanning electron microscopes (SEM), x-ray diffraction units, x-ray lithography sources, x-ray fluorescence units (e.g., Kevex and UPA Fluoroderm), cabinet radiography units (e.g., HP Faxitron and Nicolet Mikron), ion beam milling machines, and ion implanters. These radiation machines produce x-rays either intentionally or incidentally to their operation (e.g., ion implanters and ion beam milling machines). Most of these units are designed such that it is unlikely to find detectable exposures to ionizing radiation if the equipment cabinets and doors are maintained in a tight-fitting manner. Some exceptions are listed below.

During ion implantation, x-rays are formed incidental to the operation. Figure 3.11 provides a schematic diagram of a typical commercial ion implanter.[45]

Most implanters are designed with sufficient cabinet shielding (which includes lead sheeting strategically placed around the ion source housing and adjacent access doors) to maintain employee exposure below 0.25

millirems per hour (mR/hr).[46] However, some Lintott 3X implanters were found to have x-ray leakage above 2 mR/hr at the unit's surface.[47] These levels were reduced to less than 0.25 mR/hr after additional lead shielding was installed. An older Veeco ion implanter was found to have x-ray leakage around an access door (up to 1.5 mR/hr) and at a viewport (up to 0.3 mR/hr). Additional lead shielding was added to attenuate possible exposures.[48]

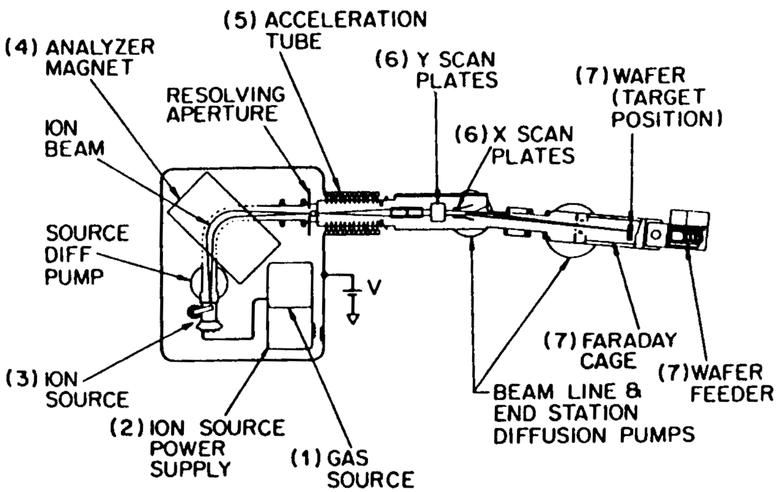

Figure 3.11. Schematic diagram of a typical commercial ion implanter.[45] *(Seidel, T. E., "Ion Implantation" in:* VLSI Technology, *© 1983, McGraw Hill, Inc. Reprinted with permission.)*

Cabinet x-ray systems are used to check the thickness of metal coatings and to identify defects (e.g., air bubbles in mold compound packages). While typically not a significant source of leakage, these units need to be checked on a periodic basis with a hand-held survey meter for x-ray leakage and to ensure door interlocks operate properly. The operator manual is also required to kept adjacent to the unit.

X-ray diffraction units generate an x-ray beam that is used to check the crystalline structure of incoming semiconductor wafers. Under certain circumstances, it is not practical to fully contain the x-ray beam in an interlocked enclosure. Operators are required to wear radiation finger badges, and controls similar to those used for a Class 4 lasers are used (e.g., enclosed room with limited access, operator training, enclosing the x-ray beam as much as practical).

One advanced technology exposure source used in lithography is x-rays. Emission levels from x-ray lithography sources may result in dose rates approaching 5 rems per year in the center of the equipment. Restricting access to areas inside the shielded wall is recommended to minimize exposure.[49]

Table 3.13 identifies potentially significant UV, RF, and ionizing radiation exposure sources published in the literature that are associated with equipment used in semiconductor manufacturing. These exposures are the maximum listed in the references cited, and should not be used as values for routine exposures.

Table 3.13. Semiconductor Manufacturing Operations with Potentially Significant Radiation Exposures. (Multiple References)

Process	Activity/Equipment	Radiation	Maximum Exposure	[Ref.]
Photolithography	UV Lamp Alignment	UV Radiation	Not Listed	[36]
X-ray Lithography	——	X-ray & Neutron	<5000 mR/yr	[49]
Etching	Plasma Etching	RF Radiation	>4.9 mW/cm^2	[36][40]
	Plasma Etcher Maint.	RF Radiation	12.9 mW/cm^2	[38][39]
Ion Implantation	Operating Lintott 3X Implanters	X-ray Radiation	>2 mR/hr	[47]
Metallization	Operating Perkin-Elmer 4400 Series Sputterers	RF Radiation	>4.9 mW/cm^2	[36]
Assembly and Test	Betascope Sources	Beta & Gamma Radiation	>5 nCi	[36]

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